

Eduardo Alejandro Aviles Jimenez

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Machine Learning Engineer with 10+ years of experience spanning full-stack development, data engineering, and AI/ML solutions across financial services, social media, and enterprise applications. Expert in building end-to-end ML pipelines, deep learning, NLP, computer vision, and predictive analytics. Proven leader in architecting scalable cloud infrastructure and DevOps practices that reduce operational costs by 30%. Strong background in technical innovation and business impact, delivering production-ready AI solutions with regulatory compliance.

SKILLS

AI/ML & LLMs	XGBoost · Scikit-learn · TensorFlow · Keras · MLflow · Pandas · DVC · Weights & Biases · LlamaIndex · LangChain · Ollama · Chainlit
Retrieval & Data	OpenSearch · PostgreSQL · Redis · RethinkDB · DynamoDB · Redshift · ETL pipelines · API integrations
Backend & Web	Python · Java · TypeScript · JavaScript · SQL · Node.js · FastAPI · NestJS · Spring Boot · Django · Flask · React · SolidJS
Cloud & MLOps	AWS ECS · Fargate · Lambda · Docker · Kubernetes · Terraform · Jenkins · CloudFormation · Ansible · GitHub Actions
Finance & Risk	IFRS 9 · Expected Credit Loss · regulatory reporting · risk modeling · loan servicing analytics

WORK EXPERIENCE

LoanPro — *Machine Learning Engineer* JUN 2022 - PRESENT

- Owned production AI/ML delivery for loan servicing and portfolio-risk workflows, translating risk policy, servicing data, and analyst needs into deployed systems.
- Led model and data architecture across risk intelligence, anomaly detection, retrieval pipelines, and conversational workflows, balancing explainability, reliability, and regulated financial requirements.
- Delivered portfolio-risk and monitoring products aligned with IFRS 9/ECL workflows, enabling earlier intervention and operational visibility for servicing teams.
- Established reusable OpenSearch/API data normalization patterns for nested loan, payment, and servicing records, improving model-readiness across multiple AI systems.
- Improved operational signal quality through anomaly detection and tenant-specific thresholding, reducing noisy events by 25% and sharpening portfolio monitoring.
- Built deployment and validation workflows across **MLflow**, **ECS**, and **Fargate** to support repeatable training, inference, monitoring, and model reliability.
- Delivered analyst-facing AI capabilities with structured outputs, retrieval grounding, and conversational interfaces, improving access to portfolio insights and recommendations.

BairesDev - HipTrain — *Senior Software Engineer* AUG 2021 - JUN 2022

- Led backend, frontend, infrastructure, and CI/CD architecture for training and scheduling platforms, improving deployment times by 30%.
- Developed core scheduling and reservation systems for training classes, improving scheduling efficiency by 80%.
- Implemented Amazon Lex chatbot workflows that automated 40% of customer inquiries and improved user interaction efficiency.

LogMeIn — *BI/ML Developer* APR 2016 - DEC 2021

- Built data integration infrastructure and ETL microservices across multiple sources, improving scalability and processing reliability.
- Implemented ML algorithms for product enhancements, improving prediction accuracy by 20%.

SSNAPP (Social Media Gateways) — *Senior Software Developer* SEP 2014 - APR 2016

- Led transition from monolithic architecture to microservices using Netflix OSS, increasing system scalability by 50% and reducing downtime by 20%.
- Optimized back-end infrastructure and implemented AWS solutions, reducing infrastructure costs by 30%.

SELECTED AI / ML PROJECTS

Risk Radar — *Loan delinquency prediction system*

LOANPRO · JAN 2025 – PRESENT

Built probability-weighted delinquency models across Current, Early/Severe Delinquency, and Default stages under IFRS 9/ECL; shipped tenant risk leaderboards, drill-down analysis, and early-warning alerts for portfolio managers and servicing teams.

XGBoostPythonPandasIFRS 9/ECLMLflow

Loan Health Intelligence — *AI-powered credit health assessment platform*

LOANPRO · FEB – APR 2025

Delivered an AI loan-risk platform with structured credit health assessments, OpenSearch and API retrieval, schema-enforced outputs, and explainable recommendations.

LlamaIndexOllamaOpenSearchChainlitPydantic

Search Analytics Copilot — *Conversational OpenSearch analytics assistant*

LOANPRO · MAY – AUG 2025

Built function-calling assistants for index discovery, metadata inspection, JSON querying, visual trend generation, loan-context retrieval, and API fallback.

LangChainLlamaIndexChainlitOpenSearchMatplotlib

Loan Monitor — *Loan creation anomaly monitoring*

LOANPRO · 2024

Designed and deployed a loan-creation monitoring system that clusters user behavior, flags anomalies, and supports per-tenant targeting decisions.

PythonMLflowScikit-learnECSFargate

LLM Copilot Lab — *Agent benchmarking and assistant frontend*

LOANPRO · SEP 2024 – JAN 2025

Benchmarked agent frameworks, model backends, embeddings, and tool-calling behavior; delivered an SSR-capable copilot frontend with Chainlit widget integration.

LangChainLlamaIndexllama-cppTypeScriptSolidJS

ML Pipeline Studio — *Churn, Rental Pricing, and Dynamic-Risk Pipelines*

PERSONAL · 2024

Built modular ML pipelines for churn prediction, automated weekly rental-price retraining, and full lifecycle management across ingestion, scoring, retraining, and deployment.

PythonMLflowCI/CDmodel monitoring

AWARDS

First Place · LoanPro Hackathon — *Risk Radar*

JAN 2025

First Place · LoanPro Hackathon — *Loan Monitoring*

APR 2024

First Place & Developers’ Choice · LogMeIn Hackathon — *LSTM churn prediction*

JUL 2018

CERTIFICATIONS – UDACITY NANODEGREES

Machine Learning DevOps Engineer

CI/CD · FastAPI · MLflow · Weights & Biases

Cloud DevOps Engineer

CloudFormation · Ansible · Jenkins · Kubernetes

Artificial Intelligence

Search · planning · pattern recognition

Deep Learning

CNNs · RNNs · GANs · sentiment analysis

EDUCATION

Computer Systems Engineering — *Engineering Degree*

JAN 2005 - SEPT 2011

Universidad De San Carlos De Guatemala

SPEAKING ENGAGEMENTS

Neural Networks from Scratch · Facebook Dev Circles Guatemala

MAR 2019

Building a working network in raw Python with no ML libraries.

RNNs and Sequence-to-Sequence Models · Facebook Dev Circles Guatemala

OCT 2018

Generating social-media threads from a Spanish-language dataset.

Neural Machine Translation · Mayan Linguistic Conference, Guatemala

JUL 2018

Applying NMT concepts toward Mayan dialect preservation.

Generative Adversarial Networks · Google IO Extended Guatemala (keynote)

MAY 2018

Facial image generation and applied generative modeling.